

Lista 1

Mec 3

$$1. H(\vec{x}, \vec{p}) = \frac{1}{2m} [\vec{p} - e \vec{A}(\vec{x})]^2 + e \Phi(\vec{x}) \quad A: \mathbb{R}^3 \rightarrow \mathbb{R}^3 \quad \Phi: \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$\vec{\pi} = \vec{p} - e \vec{A}(\vec{x}). \text{ Utilizando apenas } \{x_i, p_j\} = \delta_{ij}$$

$$\begin{aligned} x_i = \{x_i, H\} &= \{x_i, \frac{1}{2m} (p_j - e A_j)(p_j - e A_j) + e \Phi\} \\ &= \frac{1}{2m} \{x_i, (p_j - e A_j)(p_j - e A_j)\} \\ &= \frac{1}{2m} \left[(p_j - e A_j) \{x_i, p_j - e A_j\} + \{x_i, p_j - e A_j\} (p_j - e A_j) \right] \\ &= \frac{1}{m} \left[(p_j - e A_j) (\delta_{ij} - e \{x_i, A_j\}) \right] \quad (A_j = A_j(\vec{x})) \\ &= \frac{1}{m} (p_i - e A_i) \end{aligned}$$

$$\begin{aligned} p_i = \{p_i, H\} &= - \{H, p_i\} \\ &= - \left[\frac{1}{2m} \{ (p_j - e A_j)(p_j - e A_j), p_i \} + e \{ \Phi, p_i \} \right] \\ &= - \left[\frac{1}{m} (p_j + e A_j) \{ p_j - e A_j, p_i \} + e \partial_{x_i} \Phi \right] \\ &= \frac{e}{m} (p_j + e A_j) \partial_{x_i} A_j - e \partial_{x_i} \Phi \end{aligned}$$

$$\begin{aligned} 2) \{x_i, \pi_j\} &= \{x_i, p_j\} - e \{x_i, A_j\} \\ &= \{x_i, p_j\} = \delta_{ij} \end{aligned} \quad \begin{aligned} (\nabla \times \vec{A})_i &= \epsilon_{ijk} \partial_j A_k \\ \vec{p} - e \vec{A}(\vec{x}) \end{aligned}$$

$$\begin{aligned} \{\pi_i, \pi_j\} &= \{p_i - e A_i, p_j - e A_j\} = \{p_i, p_j\} - e \{p_i, A_j\} - e \{A_i, p_j\} + e \{A_i, A_j\} \\ &= -e [\{A_i, p_j\} - \{A_j, p_i\}] \\ &= e (\partial_{x_i} A_j - \partial_{x_j} A_i) \\ &= e (\epsilon_{ijk} \epsilon_{lmn} \partial_{x_m} A_n) = e \epsilon_{ijk} (\nabla \times \vec{A})_k = e \epsilon_{ijk} B_k \end{aligned}$$

b) $\dot{x}_i$ Afeito antes.

$$\frac{1}{2m} [\vec{p} - e \vec{A}(\vec{x})]^2 + e \Phi(\vec{x})$$

$$\dot{x}_i = \frac{1}{m} (\pi_i - e A_i) = \frac{\pi_i}{m}$$

$$\vec{E} = -\nabla \Phi$$

$$\dot{\pi}_i = \{ \pi_i, H \} = - \{ H, \pi_i \}$$

$$E_i = -\partial_{x_i} \Phi$$

$$= - \left[\frac{1}{2m} \{ (\vec{p} - e \vec{A}(\vec{x}))^2, \pi_i \} + e \{ \Phi, \pi_i \} \right]$$

$$= - \left[\frac{1}{2m} \{ \pi_j \pi_j, \pi_i \} + e \partial_{x_j} \Phi \{ x_j, \pi_i \} \right]$$

$$= - \left[\frac{1}{m} \pi_j \{ \pi_j, \pi_i \} + e \partial_{x_i} \Phi \right]$$

$$= - \left[\frac{e}{m} \pi_j [\partial_{x_j} A_i - \partial_{x_i} A_j] + e \partial_{x_i} \Phi \right]$$

$$= e \frac{\pi_j}{m} (\partial_{x_i} A_j - \partial_{x_j} A_i) - e \partial_{x_i} \Phi$$

$$= e [(\vec{x} \times \vec{B})_i + E_i]$$

c) $\Phi = 0$ e $\vec{B} = B_z \hat{z}$. Logo:

$\{ \pi_x, \pi_y \} = e B_z$ É a única relação.

$$X = x + \frac{\pi_y}{e B_z} \quad Y = y - \frac{\pi_x}{e B_z}$$

$$\{ X, H \} = \left\{ X, \frac{\pi_j \pi_j}{2m} \right\}$$

$$= \frac{\pi_j}{m} \left\{ x + \frac{\pi_y}{e B_z}, \pi_j \right\}$$

$$= x_j \left[\{ x, \pi_j \} + \frac{1}{e B_z} \{ \pi_y, \pi_j \} \right]$$

$$= x - \frac{1}{e B_z} x e B_z = 0 //$$

$$\{ Y, H \} = \frac{\pi_j}{m} \left\{ y - \frac{\pi_x}{e B_z}, \pi_j \right\}$$

$$= x_j \left[\{ y, \pi_j \} - \frac{1}{e B_z} \{ \pi_x, \pi_j \} \right]$$

$$= y - \frac{1}{e B_z} y e B_z = 0 //$$

$$\{ X, Y \} = \left\{ x + \frac{\pi_y}{e B_z}, y - \frac{\pi_x}{e B_z} \right\}$$

$$= - \left\{ x, \frac{\pi_x}{e B_z} \right\} - \left\{ y, \frac{\pi_y}{e B_z} \right\} - \left\{ \frac{\pi_y}{e B_z}, \frac{\pi_x}{e B_z} \right\}$$

$$= - \frac{2}{e B_z} + \frac{1}{e^2 B_z^2} e B_z = - \frac{1}{e B_z} \quad \text{Já que não é nulo, } X \text{ muda em relação ao fluxo de } X.$$

$$d) \quad \pi_x = (X-x)eB \quad \pi_y = (y-Y)eB$$

$$H_{\perp} = \frac{(x-X)^2 e^2 B^2}{2m} + \frac{(y-Y)^2 e^2 B^2}{2m}$$

$$q = x - X \quad p = eB(y - Y)$$

$$\{q, p\} = eB \{x - X, y - Y\}$$

$$= eB (\{x, y\} - \{x, Y\} - \{X, y\} + \{X, Y\})$$

$$= eB (-\{x, y - \frac{\pi_x}{eB}\} - \{x + \frac{\pi_y}{eB}, y\} - \frac{1}{eB})$$

$$= 1 + 1 - 1 = 1$$

$$H_{\perp} = \frac{q^2}{2m} e^2 B^2 + \frac{p^2}{2m} = \frac{p^2}{2m} + \frac{1}{2} \frac{e^2 B^2}{m} q^2 \Rightarrow \omega_c = \frac{eB}{m} \leftarrow \text{errado}$$

se B negativo, $\omega_c = -\frac{eB}{m}$ por conta da raiz tomada

$$2. H = \frac{p^2}{2m} - \frac{\alpha}{r}$$

$$\vec{L} = \vec{r} \times \vec{p} \Rightarrow L_i = \epsilon_{ijk} x_j p_k$$

$$\{L_i, x_j\} = \epsilon_{ikm} x_k \{p_m, x_j\} = \epsilon_{ijk} x_k$$

$$\{L_i, p_j\} = \epsilon_{ilk} p_k \{x_l, p_j\} = \epsilon_{ijk} p_k$$

$$\{x_i, H\} = \{x_i, \frac{p^2}{2m}\} = -\frac{p_i}{m} = -\dot{x}_i$$

$$\begin{aligned} \{p_i, H\} &= \alpha \left\{ \frac{1}{r}, p_i \right\} = -\alpha \frac{1}{r^2} \{r, p_i\} = -\frac{\alpha}{r^2} \{(\underline{x} \cdot \underline{x})^{\frac{1}{2}}, p_i\} \\ &= -\frac{\alpha}{r^2} \frac{1}{2} \underbrace{(x_j x_j)^{-\frac{1}{2}}}_{r^{-1}} \{x^2, p_i\} = -\frac{\alpha}{r^3} x_i = -\dot{p}_i \end{aligned}$$

$$\vec{A} = \vec{p} \times \vec{L} - m\alpha \hat{r}$$

$$A_i = \epsilon_{ijk} p_j L_k - m\alpha \frac{x_i}{r}$$

$$= \epsilon_{ijk} p_j \epsilon_{kmn} x_m p_n - m\alpha \frac{x_i}{r} = x_i p_j p_j - x_j p_j p_i - m\alpha \frac{x_i}{r}$$

$$= (p^2 - \frac{m\alpha}{r}) x_i - (\vec{r} \cdot \vec{p}) p_i$$

$$\vec{A} = (r^2 - \frac{m\alpha}{r}) \vec{r} - (\vec{r} \cdot \vec{p}) \vec{p}$$

$$2) \{L_i, H\} = \epsilon_{ijk} \{x_j p_k, H\} = \epsilon_{ijk} (\{x_j, H\} p_k + x_j \{p_k, H\})$$

$$= \epsilon_{ijk} \left(\frac{p_j}{m} p_k - \frac{\alpha}{r^3} x_k x_j \right) = \epsilon_{ikj} \left(\frac{p_k}{m} p_j - \frac{\alpha}{r^3} x_j x_k \right)$$

$$\begin{aligned} &\xrightarrow{\quad} = -\epsilon_{ijk} \left(\frac{p_j}{m} p_k - \frac{\alpha}{r^3} x_k x_j \right) \\ &\xrightarrow{\quad} = 0_k \end{aligned}$$

$\{x_j p_k, H\}$ é simétrico, ϵ_{ijk} anti.

$$\{A_i, H\} = \epsilon_{ijk} \{p_j L_k, H\} - m\alpha \left\{ \frac{x_i}{r}, H \right\}$$

$$= \epsilon_{ijk} L_k \underbrace{\{p_j, H\}}_{=0} - m\alpha \left\{ \frac{x_i}{r}, \frac{p^2}{2m} \right\}$$

$$= -\epsilon_{ijk} L_k \frac{\alpha}{r^3} x_j - \frac{m\alpha}{2} \left[x_i \left\{ \frac{1}{r}, \frac{p^2}{2m} \right\} + \frac{1}{r} \left\{ x_i, \frac{p^2}{2m} \right\} \right]$$

$$= -\epsilon_{ijk} \epsilon_{kmn} x_m p_n \frac{\alpha}{r^3} x_j - \alpha \left[x_i p_j \left\{ \frac{1}{r}, p_j \right\} + \frac{1}{r} p_j \{x_i, p_j\} \right]$$

$$= - \left[x_i p_j x_j \frac{\alpha}{r^3} - x_j p_i x_j \frac{\alpha}{r^3} - \frac{\alpha}{r^2} x_i p_j \{r, p_j\} + \frac{\alpha}{r} p_i \right]$$

$$= - \left[u_i p_j u_j \frac{\alpha}{r^3} - u_j p_i u_j \frac{\alpha}{r^3} - \frac{\alpha}{r^2} u_i p_j \{r, p_j\} + \frac{\alpha}{r} p_i \right]$$

$$= - \left[u_i p_j u_j \frac{\alpha}{r^3} - \overbrace{u_j p_i u_j}^{=r^2} \frac{\alpha}{r^3} - \frac{\alpha}{r^3} u_i p_j u_j + \frac{\alpha}{r} p_i \right]$$

$$= - \left[\frac{\alpha}{r} p_i - \frac{\alpha}{r} p_i \right] = 0$$

$$\begin{aligned}
b) \{L_i, L_j\} &= \epsilon_{imn} \epsilon_{jkl} \{x_m p_n, x_k p_l\} \\
&= \epsilon_{imn} \epsilon_{jkl} \left[x_k \{x_m, p_l\} p_n + p_l \{p_n, x_k\} x_m \right] \\
&= \epsilon_{imn} \epsilon_{jkl} x_k p_n - \epsilon_{imk} \epsilon_{jkl} p_l x_m \\
&= \epsilon_{imk} \epsilon_{kjl} p_l x_m - \epsilon_{imn} \epsilon_{msk} x_k p_n \\
&= p_m x_n - p_i x_j - x_k p_k + x_i p_i \\
&= x_i p_j - x_j p_i \\
&= \epsilon_{ijk} (\epsilon_{kmn} x_m p_n) = \epsilon_{ijk} L_k
\end{aligned}$$

$$\begin{aligned}
\{L_i, A_j\} &= \{L_i, \epsilon_{jmn} p_n L_m\} - \{L_i, m \alpha \frac{x_j}{r}\} \\
&= \epsilon_{jmn} \{L_i, p_m\} L_n + \epsilon_{jmn} \{L_i, L_n\} p_m - m \frac{\alpha}{r} \{L_i, x_j\} - m \alpha x_j \{L_i, \frac{1}{r}\} \\
&= \epsilon_{jmn} \epsilon_{ikl} p_l \{x_k, p_m\} L_n + \epsilon_{jmn} \epsilon_{ikl} L_k p_m - m \frac{\alpha}{r} \epsilon_{imn} x_m \{p_n, x_j\} + m x_j \frac{\alpha}{r^2} \{L_i, r\} \\
&= \epsilon_{jnm} \epsilon_{nikl} p_l L_n - \epsilon_{jmn} \epsilon_{nikl} L_k p_m - m \frac{\alpha}{r} \epsilon_{imj} x_m + m x_j \frac{\alpha}{r^2} x_k \{L_i, x_k\} \\
&= p_k L_k - p_j L_i - p_k L_k + p_i L_j - m \frac{\alpha}{r} \epsilon_{ijk} x_k + m x_j \frac{\alpha}{r^2} x_k \epsilon_{imn} x_m \{p_n, x_k\} \\
&= p_i L_j - p_j L_i - m \frac{\alpha}{r} \epsilon_{ijk} - \underbrace{\epsilon_{imn}}_{\text{anti}} m \frac{\alpha}{r^2} \underbrace{x_j x_m x_n}_{\text{symmetric}} \xrightarrow{0} \\
&= \epsilon_{ijk} \left[\epsilon_{kmn} p_i L_j - m \frac{\alpha}{r} \right] \\
&= \epsilon_{ijk} A_k
\end{aligned}$$

$\epsilon_{ijk} p_k \{L_i, p_j\}$

$$\begin{aligned}
\{A_i, p_j\} &= p_m \epsilon_{imn} \{L_n, p_j\} - m \alpha \left\{ \frac{x_i}{r}, p_j \right\} \\
&= p_m \epsilon_{imn} \epsilon_{njk} p_k - m \alpha \left(x_i \left\{ \frac{1}{r}, p_j \right\} + \frac{1}{r} \{x_i, p_j\} \right) \\
&= p_k p_k - p_i p_j + m \frac{\alpha}{r^2} x_i x_j + m \frac{\alpha}{r} \delta_{ij}
\end{aligned}$$

$$\{A_i, A_j\} = \left\{ \epsilon_{imn} p_m L_n - m \alpha \frac{x_i}{r}, A_j \right\}$$

$$= \epsilon_{imn} \{p_m L_n, A_j\} - m \alpha \left\{ \frac{x_i}{r}, A_j \right\}$$

$$= \epsilon_{imn} (p_m \{L_n, A_j\} + L_n \{p_m, A_j\}) - m \alpha x_i \left\{ \frac{1}{r}, A_j \right\} - m \alpha \frac{1}{r} \{x_i, A_j\}$$

$$= \epsilon_{imn} \left[p_m \epsilon_{nj k} A_k - L_n (\epsilon_{j k l} p_k \{L_l, p_m\} - m \alpha \left\{ \frac{x_j}{r}, p_m \right\}) \right] + m x_i \frac{\alpha}{r^3} x_k \{x_k, A_j\} - m \alpha \frac{1}{r} \{x_i, A_j\}$$

$$\{x_i, A_j\} = \epsilon_{jmn} \{x_i, p_m L_n\} = \epsilon_{jmn} (\{x_i, p_m\} L_n + p_m \{x_i, L_n\})$$

$$= \epsilon_{j i n} L_n - \epsilon_{j m n} \epsilon_{n i k} x_k p_m = \epsilon_{j i n} L_n - x_k p_k + x_j p_i$$

$$= x_j p_i - x_k p_k - \epsilon_{i j k} L_k$$

$$x_i \{x_i, A_j\} = x_i \epsilon_{j i n} L_n - \epsilon_{j m n} \epsilon_{n i k} x_i x_k p_m$$

$$= - (x_j x_m p_m - x_m x_j p_m) - x_i \epsilon_{i j k} L_k$$

$$= -x_i \epsilon_{i j k} L_k$$

$$\dots = \epsilon_{imn} \epsilon_{n j k} p_m A_k - \epsilon_{imn} L_n \left[\epsilon_{j k l} p_k \epsilon_{l m o} p_o - m \alpha (x_j \left\{ \frac{1}{r}, p_m \right\} + \frac{1}{r} \{x_j, p_m\}) \right]$$

$$- m x_i \frac{\alpha}{r^3} x_k \epsilon_{k j n} L_n - m \alpha \frac{1}{r} (x_j p_i - x_k p_k - \epsilon_{i j k} L_k)$$

$$= p_k A_k - p_j A_i - L_n \epsilon_{i n m} \epsilon_{m l o} \epsilon_{j k l} p_k p_o + \epsilon_{i m n} L_n m \alpha (-\frac{x_j}{r^3} x_k \delta_{k m} + \frac{1}{r} \delta_{j m}) + (\dots)$$

$$= p_k \epsilon_{i k m n} p_m L_n - p_k m \alpha \frac{x_k}{r} - p_j \epsilon_{i m n} p_m L_n + p_j m \alpha \frac{x_i}{r} - L_n \epsilon_{j k i} p_k p_n + L_n \epsilon_{j k n} p_k p_i$$

$$- \epsilon_{i m n} L_n m \frac{\alpha}{r^3} x_j x_m + \epsilon_{i j k} L_k m \frac{\alpha}{r}$$

$$+ m x_i \frac{\alpha}{r^3} x_k x_j p_k - m x_i \frac{\alpha}{r^3} x_k \epsilon_{k j n} L_n - \epsilon_{k j m} L_m m x_i \frac{\alpha}{r^3} x_k$$

$$- m \frac{\alpha}{r} x_j p_i + m \frac{\alpha}{r} x_k p_k + m \frac{\alpha}{r} \epsilon_{i j k} L_k$$

$$= -p_j \epsilon_{i m n} p_m L_n + m \alpha r (x_i p_j - x_j p_i) + L_n p_k (\epsilon_{j k n} p_i - \epsilon_{i j k} p_n)$$

$$- m \frac{\alpha}{r^3} (\epsilon_{i m n} x_j x_m L_n + \epsilon_{k j m} x_i x_k L_m) + 2 m \frac{\alpha}{r} \epsilon_{i j k} L_k$$

$$+ m x_i \frac{\alpha}{r^3} (x_j x_k p_k - x_k \epsilon_{i k n} L_n)$$

$$= m \frac{\sigma}{r} (\epsilon_{ijk} \overbrace{\epsilon_{kmn} x_m p_n}^{L_k}) + L_n p_k (\epsilon_{ijk} p_i - p_j \epsilon_{ikn} - \epsilon_{jln} p_k)$$

$$+ m \frac{\sigma}{r^3} (\epsilon_{cmn} x_j x_m L_n + \epsilon_{kjm} x_i x_k L_m + x_i x_k p_k x_j - \epsilon_{kjm} x_i x_k L_n)$$

$$+ 2 m \frac{\sigma}{r} \epsilon_{ijk} L_k$$

$$= 3 m \frac{\sigma}{r} \epsilon_{ijk} L_k + m \frac{\sigma}{r^3} [x_i x_k (\epsilon_{kjm} L_m - \epsilon_{kjm} L_n) + \epsilon_{imn} x_j x_m L_n + x_i x_k p_k x_j]$$

$$+ L_n p_k (\epsilon_{ijk} p_i - p_j \epsilon_{ikn})$$

$$\hookrightarrow = -\epsilon_{nim} \epsilon_{mqp} \epsilon_{pjk} p_q$$

$$= -\epsilon_{nim} (\delta_{jm} \delta_{kq} - \delta_{jq} \delta_{km}) p_q$$

$$= -\epsilon_{nij} p_k + \epsilon_{nik} p_j$$

$$- L_n p_k (\epsilon_{nij} p_k + \epsilon_{nik} p_j - p_j \epsilon_{ikn}) = -p^2 \epsilon_{ijk} L_k$$

$$\begin{aligned} \epsilon_{imn} x_j x_m L_n &= \epsilon_{imn} x_j x_m \epsilon_{nop} x_o p_p \\ &= x_j x_m x_i p_m - x_j x_m x_m p_i \\ &= x_{ik} p_k x_j x_i - r^2 x_j p_i \end{aligned}$$

$$\dots = 3 m \frac{\sigma}{r} \epsilon_{ijk} L_k - m \frac{\sigma}{r} (x_j p_i - x_i p_j) + m \frac{\sigma}{r^3} (x_{ik} p_k x_j x_i - x_i x_k p_k x_j) - p^2 \epsilon_{ijk} L_k$$

$$= 3 m \frac{\sigma}{r} \epsilon_{ijk} L_k - p^2 \epsilon_{ijk} L_k - m \frac{\sigma}{r} (x_j p_i - x_i p_j)$$

$$\epsilon_{ijk} \epsilon_{kmn} x_m p_n = \epsilon_{ijk} L_k$$

$$\{A_i, A_j\} = (2 m \frac{\sigma}{r} - p^2) \epsilon_{ijk} L_k = -2 m H \epsilon_{ijk} L_k \quad \square$$

$$\begin{aligned}
 A_i L_i &= \left(\epsilon_{jki} p_j L_k - m \alpha \frac{x_i}{r} \right) \left(\epsilon_{lmn} x_m p_n \right) \\
 &= p_j L_k x_k p_i - p_j L_k x_i p_k - \underbrace{\left(\epsilon_{lmn} x_m p_n m \alpha \frac{x_i}{r} \right)}_{\rightarrow 0} \quad \text{Anti-simétrico em [mi]} \\
 &= \epsilon_{kmn} x_m p_n p_j x_k p_i - \epsilon_{kmn} x_m p_n p_j x_i p_k = 0 \quad \text{Anti-simétrico em [km]} \\
 &\quad \text{e [kn] respectivamente}
 \end{aligned}$$

$$\begin{aligned}
 A_i A_i &= \left(\epsilon_{jki} p_j L_k - m \alpha \frac{x_i}{r} \right) \left(\epsilon_{lmn} p_l L_n - m \alpha \frac{x_i}{r} \right) \\
 &= \epsilon_{jki} \epsilon_{ilm} p_j L_k p_l L_n - 2 m \alpha \frac{x_i}{r} \epsilon_{jki} p_j L_k + m^2 \alpha^2 \frac{x_i^2}{r^2} \\
 &= p^2 L^2 - p_j L_k p_k L_j - 2 m \alpha \frac{x_i}{r} \epsilon_{ijl} p_j L_l + m^2 \alpha^2 \\
 &= p^2 L^2 - \underbrace{\epsilon_{kmn} x_m p_n p_j p_k L_j}_{\rightarrow 0 \text{ (nm)}} - 2 m \alpha \frac{x_i}{r} \epsilon_{kij} x_i p_j L_k + m^2 \alpha^2 \\
 &= \left(p^2 - 2 m \alpha \frac{x_i}{r} \right) L^2 + m^2 \alpha^2 //
 \end{aligned}$$

$$d) D_i = \frac{A_i}{\sqrt{2mH}} \quad J_{\pm i} = \frac{1}{2}(L_i \pm D_i) \quad \{L_i \pm D_i, L_i \mp D_i\}$$

$$\begin{aligned}
 \{J_{\pm i}, J_{\pm j}\} &= \frac{1}{4} \{L_i \pm D_i, L_j \pm D_j\} \\
 &= \frac{1}{4} (\{L_i, L_j\} \pm \{L_i, D_j\} \pm \{D_i, L_j\} \pm \{D_i, D_j\})
 \end{aligned}$$

$$\{L_i, D_j\} = \frac{1}{\sqrt{2mH}} \{L_i, A_j\} = \frac{1}{\sqrt{2mH}} \epsilon_{ijl} A_l$$

$$\begin{aligned}
 \{J_{\pm i}, J_{\pm j}\} &= \frac{1}{4} \left(\epsilon_{ijl} L_l \pm \frac{1}{\sqrt{2mH}} \epsilon_{ijl} A_l \mp \frac{1}{\sqrt{2mH}} \epsilon_{jil} A_l \pm \frac{1}{\sqrt{2mH}} \epsilon_{ijl} L_l \right) \\
 &= \frac{1}{2} (\epsilon_{ijl} L_l \pm \epsilon_{ijl} A_l) \\
 &= \epsilon_{ijl} J_{\pm l}
 \end{aligned}$$

$$\begin{aligned}
 \{J_{+i}, J_{-j}\} &= \frac{1}{4} \left(\epsilon_{ijl} L_l - \frac{1}{\sqrt{2mH}} \epsilon_{ijl} A_l - \frac{1}{\sqrt{2mH}} \epsilon_{jil} A_l - \epsilon_{ijl} L_l \right) \\
 &= 0 //
 \end{aligned}$$